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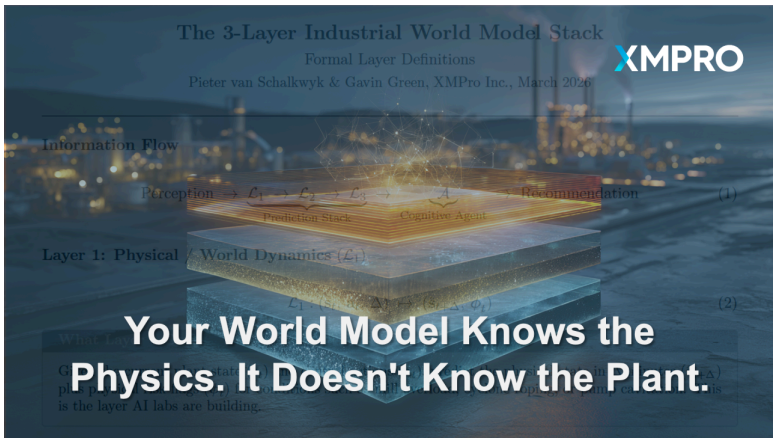

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Your World Model Knows the Physics. It Doesn't Know the Plant.

Pieter van Schalkwyk
Industrial Operations & Engineering | Defining the right problems before automation and AI

March 18, 2026

Two conversations are happening in industrial AI right now. They are not yet connected.

The first is about **world models**. These are AI systems that learn how physical environments change under actions. AMI Labs (founded by **Yann LeCun**) raised over $1 billion to build them. Meta's V-JEPA 2 showed that self-supervised video models can learn physical reasoning. World Labs is building persistent 3D scene models that respond to actions.

The second is about **cognitive AI agents**. These are multi-agent systems that observe, reflect, plan, and act in complex environments. They reason about what to do, not just what will happen. Industry is deploying them for operations, maintenance, safety, and scheduling decisions.

How do these fit together? Are world models and cognitive agents competing approaches? Or are they complementary parts of the same architecture?

I believe they are complementary. Understanding how they compose is the key to making either one work in production.

The Problem with Each Approach Alone

World models without agents are prediction engines with no one to use the predictions. A world model can show how particle size distribution changes when mill speed increases in mining, for example. That captures the physical cause-and-effect relationship.

But a world model cannot decide that mill speed should change in the first place. It cannot evaluate whether the downstream KPI impact is worth it. It cannot check whether the resulting alarm rate will overwhelm the operator. It cannot explain the trade-off reasoning or learn from the operator's response. *Prediction without reasoning is a dashboard, not autonomy.*

Cognitive agents without world models are like a *new graduate engineer* arriving at a plant. They have textbook knowledge and can run the calculations. They can observe plant state, reason about what actions might improve throughput, and explain their logic. They can compute energy balances, apply Bond's Law, or solve material balance equations using engineering libraries as tools.

Cognitive agents also have memory and learning built in. Over time, they accumulate operational experience from every planning cycle, every operator override, and every predicted-versus-actual comparison. But building that experience takes time. A new engineer needs years on site before their instincts match the veteran operator who knows exactly how *this* mill responds when the ore hardens on a Tuesday night shift.

World models compress that learning curve. They encode real plant experience, learned from operational data, capturing the non-linearities, interactions, and site-specific behaviours that textbook equations approximate but do not fully describe. ***A cognitive agent with a calibrated world model is closer to that veteran operator from day one.*** It still learns and improves with experience, but it starts from a much higher baseline of plant-specific knowledge.

The architecture that works is agents using world models as tools. Agents augment their existing engineering and reasoning capabilities with calibrated, plant-specific prediction. This makes their recommendations both well-reasoned and grounded in how the real plant actually behaves.

One Layer of World Model Is Not Enough

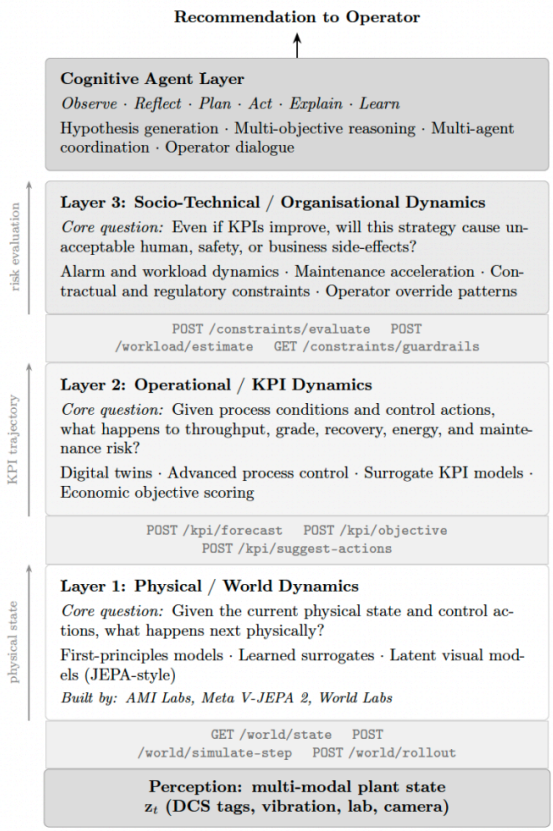
This is where current world model programmes fall short for industrial use. Every major programme (JEPA, V-JEPA 2, Dreamer, MuZero, Marble) models the same thing: physical and spatial dynamics. How materials behave. How objects move. How scenes change under actions. Call this **Layer 1**, the physics layer. Industrial AI has always had textbook engineering and physics math. What it has not had is learned models that capture how a specific plant actually responds, including the real-world variability, wear patterns, and process interactions that first-principles equations approximate but cannot fully predict. That is what world models add.

Industrial autonomy requires understanding at three levels, not one.

An AI that understands how particle size distributions change when you increase mill speed is valuable. Build on that an AI that predicts

what that change does to throughput, concentrate grade, energy cost, and recovery over the next two hours. Now add an AI that knows the proposed strategy will triple the alarm rate, accelerate liner wear by 15%, violate a contractual shipment specification, and overwhelm the night-shift operator.

The first understands physics. The second understands operations. The third understands the organisation. You need all three for a deployable system.



3 Layer Agentic Operations Stack (see link to paper at the end of this article)

The industrial world model must be a **3-layer stack**.

Layer 1: Physical / World Dynamics

Core question: *Given the current physical state and control actions, what happens next physically?*

This is what AI labs are building. For industry, Layer 1 includes first-principles models (breakage matrices, population balance equations, fluid dynamics), learned surrogates trained from simulation and historian data, and vision-based latent models. These latent models understand states like mill overload or poor froth stability from camera feeds. AI labs are investing substantially here. This is the foundation everything else builds on.

Layer 2: Operational / KPI Dynamics

Core question: *Given process conditions and control actions, what happens to throughput, grade, recovery, energy, and maintenance risk?*

Layer 2 translates physical state changes into operational value. Digital twins, advanced process control, and reinforcement learning operate at this level. Pieces of Layer 2 exist in scattered form: Emerson Mimic, ABB/Boliden Dymola models, bespoke APC systems. But they are not built as composable prediction services that agents can query programmatically. They were built for human dashboards and control room screens.

Layer 3: Socio-Technical / Organisational Dynamics

- **Core question:** *Even if KPIs improve, will this operating strategy cause unacceptable human, safety, or business side-effects?*

This layer rarely appears in the world model literature. It is domain-specific rather than foundational. But it determines whether autonomous action is acceptable in real operations:

- **Alarm and workload dynamics.** A control strategy that reduces KPI variance might triple the alarm rate during ore hardness transitions.
- **Maintenance acceleration.** Running at aggressive throughput for 12 hours may trigger accelerated equipment failure that wipes out a week of gains.
- **Business constraints.** Contractual penalties, shipment specifications, insurance triggers, environmental reporting thresholds.
- **Organisational signals.** Persistent operator overrides are data. They indicate unmodelled constraints that no physics model captures.

How Cognitive Agents Use the 3-Layer Stack

This is where the two conversations connect.

Cognitive agents use the 3-layer world model stack as their grounding mechanism. The world model stack gives agents calibrated foresight. The agents provide the reasoning, hypothesis generation, and explanation that prediction services cannot.

Consider a cognitive agent tasked with increasing throughput by 5% on a copper grinding-flotation circuit. Here is how the observe-reflect-plan-act cycle works with the world model stack:

- **Observe** the current plant state across all three layers. Physical conditions, KPI trajectory, and active constraints.
- **Reflect** on what the current state means in operational context. Is the ore hardening? Is a shift change approaching? Are we near a contractual shipment threshold?
- **Plan** by generating diverse candidate strategies. Not just perturbing setpoints, but reasoning about fundamentally different approaches to the goal. Then simulate each candidate through Layer 1 (physical outcomes), Layer 2 (KPI impacts), and Layer 3 (socio-technical risk).
- **Act** by presenting ranked recommendations to the operator with full explanations. Not just "do this" but why this plan is preferred, what trade-offs were considered, what could go wrong, and

what the agent would recommend if the operator modifies a constraint.

Cognitive agents can do all of this without world models. They can run engineering calculations, apply process models, and reason about constraints. But their predictions are based on idealised models, not on how this specific plant is represented today. The world model stack closes that gap. *Layer 1* calibrates the agent's physical predictions against real plant dynamics. *Layer 2* grounds its KPI forecasts in actual operational data rather than textbook relationships. *Layer 3* captures the site-specific constraints, alarm patterns, and organisational behaviours that no engineering equation models.

The result is not a new capability. It is a step change in *reliability*. The agent reasons the same way, but its predictions are calibrated against plant reality rather than textbook theory.

World models improve the reliability and predictability of agentic operations. They are what moves cognitive agents from textbook-grade predictions to plant-grade predictions, and that difference determines whether operators trust the system enough to act on its recommendations.

Progressive Autonomy

The combination of cognitive agents and the 3-layer world model stack enables a natural progression from passive to active.

- **Shadow Mode.** Agents run in parallel with existing control. They use the world model stack to predict what they would have recommended. Humans retain full control. This phase calibrates the world models and builds trust.
- **Advisory Mode.** Agents generate recommendations with simulated outcomes through all three layers. Operators approve, modify, or reject each recommendation. Each override teaches the system about constraints it had not captured.
- **Bounded Autonomous.** Low-risk actions within defined envelopes execute autonomously. The system escalates high-risk or out-of-envelope situations. The envelope expands based on demonstrated prediction accuracy.
- **Collaborative Autonomous.** The system manages routine operations. Humans focus on exceptions, strategic decisions, and boundary management.

Each phase generates the calibration data the next phase requires. You cannot skip steps. The world model stack is what makes each transition possible. You cannot grant an agent more authority without demonstrated prediction accuracy across all three layers.

The Opportunity

Two developments are converging. AI labs are building physical world models (Layer 1). Multi-agent cognitive systems are maturing for industrial decision-making. The missing piece is the composition: a 3-layer world model stack that gives cognitive agents calibrated, physics-grounded foresight across physical, operational, and socio-technical dimensions.

Layer 1 provides the physical foundation that AI labs are building. Layers 2 and 3 encode the operational and organisational intelligence that industry has accumulated but not yet exposed as composable prediction services. Cognitive agents provide the reasoning layer that translates calibrated prediction into action.

The 3-layer stack is how world models and cognitive agents come together. It is how agentic operations move from demonstrations to production systems that create measurable industrial value, safely and at scale.

The Full Technical Framework

This article introduces the concept. For the full architectural specification, see our companion paper:

"The 3-Layer Industrial World Model Stack: Why AI Labs Are Building One-Third of What Industrial Autonomy Requires" *Pieter van Schalkwyk & Gavin Green, XMPro Inc., March 2026*

Link to Paper: **https://github.com/XMPro/Multi-Agent/blob/main/research/2026.03.19_Three_Layer_World_Model_for_Industrial_Operations.pdf**

The 3-Layer Industrial World Model Stack

Formal Layer Definitions

Pieter van Schalkwyk & Gavin Green, XMPro Inc., March 2026

Information Flow

$$\text{Perception} \rightarrow \underbrace{\mathcal{L}_1 \rightarrow \mathcal{L}_2 \rightarrow \mathcal{L}_3}_{\text{Prediction Stack}} \rightarrow \underbrace{\mathcal{A}}_{\text{Cognitive Agent}} \rightarrow \text{Recommendation} \tag{1}$$

Layer 1: Physical / World Dynamics ($\mathcal{L}_1$)

$$\mathcal{L}_1 : (\mathbf{s}_t,\, \mathbf{a}_t,\, \Delta) \;\mapsto\; (\hat{\mathbf{s}}_{t+\Delta},\, \phi_t) \tag{2}$$

What Layer 1 Does

Given the current plant state ($\mathbf{s}_t$) and control actions ($\mathbf{a}_t$), predict the physical state in Δ minutes ($\hat{\mathbf{s}}_{t+\Delta}$) plus physical risk flags (ϕ_t) for conditions such as mill overload, cyclone roping, or pump cavitation. This is the layer AI labs are building.

Layer 2: Operational KPI Dynamics ($\mathcal{L}_2$)

$$\mathcal{L}_2 : (\{\hat{\mathbf{s}}\}_t^{t+H},\, \mathbf{a}_{t:t+H},\, H) \;\mapsto\; (\hat{\mathbf{k}}_{t:t+H},\, v,\, \sigma_k) \tag{3}$$

What Layer 2 Does

Given the physical state trajectory from Layer 1, predict KPI forecasts ($\hat{\mathbf{k}}$) for throughput, grade, recovery, energy, and maintenance risk. Returns a single economic score (v) and a confidence estimate (σ_k). This layer turns physics into business value.

Layer 3: Socio-Technical Dynamics ($\mathcal{L}_3$)

$$\mathcal{L}_3 : (\hat{\mathbf{k}}_{t:t+H},\, \mathbf{a}_{t:t+H},\, \mathcal{C}) \;\mapsto\; (\rho,\, \mathbf{F},\, \omega,\, E,\, \Delta\mathcal{C}) \tag{4}$$

What Layer 3 Does

Given KPI trajectories and a proposed action schedule, evaluate against the constraint knowledge base ($\mathcal{C}$). Returns:

- $\rho \in [0, 1]$: overall risk score (0 = no concern, 1 = unacceptable)
- $\mathbf{F}$: binary safety and compliance flags (one per constraint)
- ω: operator workload estimate (alarm rate, cognitive load)
- E: natural language explanation of the risk assessment
- $\Delta\mathcal{C}$: proposed updates to the constraint knowledge base from observed operator behaviour

E requires the system to reason in language about risk. $\Delta\mathcal{C}$ requires the system to learn new constraints from operational experience. Both define the technology needed at Layer 3.

From: "The 3-Layer Industrial World Model Stack" (van Schalkwyk & Green, 2026)

The paper covers formal layer definitions, API boundaries, the knowledge acquisition problem, operator co-evolution requirements, cognitive agent

specifications, and progressive autonomy gate criteria. It includes end-to-end demonstrations on both a copper grinding-flotation circuit and an upstream oil and gas separation train. It also maps the architecture to LeCun's AMI blueprint.

Pieter van Schalkwyk is the CEO of XMPro, specializing in industrial AI agent orchestration and governance. Drawing on 30+ years of experience in industrial automation, he helps organizations implement practical AI solutions that deliver measurable business outcomes while ensuring responsible AI deployment at scale.

*He authored the **Industrial AI Agent Manifesto**, published by the Digital Twin Consortium in February 2026, a governance framework for trustworthy autonomous operations*

*Our **GitHub Repo** has more technical information. You can also contact myself or **Gavin Green** for more information.*

*Read more on MAGS at **The Digital Engineer***

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Love the Shadow Mode, that can act as an Expert System and be the bridge between the World Model and and the Cognitive AI **Agent.Do** you some actual use Cases for the 3_Layer Model that you have implemented.

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Hezrone Good question. We have production implementations of all three layers in customer solutions, with our MAGS cognitive agents operating across all of them. Layers 2 and 3 (KPI dynamics and socio-technical constraints) are operational. Layer 1 currently includes engineering and physics calculations that agents u ...more

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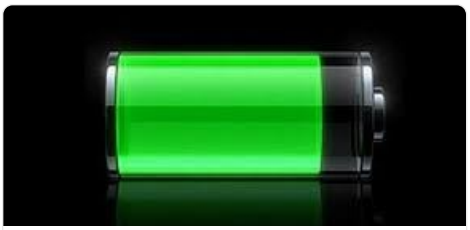

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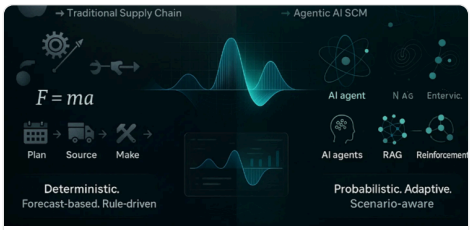

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